DRUGS · CORTICOSTEROIDS

Dexamethasone

Dexamethasone is a potent, long-acting glucocorticoid with almost no mineralocorticoid effect. In ENT it is given intravenously or orally for airway oedema, croup and angioedema, in a single dose at tonsillectomy to reduce nausea and pain, at induction for many head and neck operations, with antibiotics in bacterial meningitis to protect hearing, and through the drum for sudden hearing loss and Ménière's disease.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

DEXAMETHASONE

	SUMMARY
Class	Long-acting glucocorticoid (about 25 times the anti-inflammatory potency of hydrocortisone, biological half-life 36 to 54 hours)
How it works	Glucocorticoid receptor agonist; no salt retention
ENT uses	Croup; airway oedema before extubation and after airway surgery; angioedema (allergic, as an adjunct); peritonsillar abscess (single dose); tonsillectomy (nausea and pain); post-operative oedema after facial and thyroid surgery; bacterial meningitis (with the first antibiotic); intratympanic treatment; cerebral oedema
Adult dose	Airway oedema commonly 6 to 8 mg intravenously every 8 hours for 2 to 3 doses; nausea prophylaxis 4 to 8 mg at induction; quinsy 8 to 10 mg once; meningitis 10 mg every 6 hours for 4 days
Child dose	Croup 150 micrograms/kg orally once (up to 600 micrograms/kg in severe croup); tonsillectomy 150 micrograms/kg intravenously (maximum 8 to 10 mg); meningitis 150 micrograms/kg every 6 hours for 4 days
Pregnancy	Crosses the placenta (unlike prednisolone); use when indicated

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

It reduces capillary permeability, mucosal oedema and cytokine release. Its long action means one dose covers the risk period of a tonsillectomy or an extubation. Its central antiemetic effect is not fully understood.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Croup and the airway

A single oral dose of dexamethasone for **croup** reduces return visits, admissions and intubation, even in mild disease; nebulised adrenaline is added for severe stridor. Before extubation after prolonged intubation or airway surgery, steroids reduce post-extubation stridor in high-risk patients.

3.2 Tonsillectomy

A single intravenous dose reduces post-operative vomiting and improves early oral intake, and is recommended by the AAO-HNS guideline; it has not been convincingly shown to increase bleeding.

3.3 Bacterial meningitis

Dexamethasone given **just before or with the first antibiotic dose** reduces hearing loss and neurological sequelae in pneumococcal and Haemophilus meningitis; it is not started more than about 4 to 12 hours after antibiotics.

3.4 Intratympanic use

Dexamethasone 10 to 24 mg/ml is injected through the drum for sudden hearing loss and Ménière's disease (see intratympanic steroids).

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Hyperglycaemia (even after a single dose in diabetics), insomnia, agitation, perineal burning or itch with rapid intravenous injection.
- Long courses carry all glucocorticoid harms, with marked adrenal suppression.
- Tumour lysis syndrome in undiagnosed lymphoma: avoid steroids before tissue diagnosis when lymphoma is suspected (they can also make the biopsy non-diagnostic).

4.2 Interactions

Enzyme inducers (rifampicin, phenytoin, carbamazepine) reduce its effect; it reduces the effect of some drugs metabolised by CYP3A4.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Dexamethasone in **croup**: dose and evidence.
- Timing in **bacterial meningitis** and why ENT surgeons care (hearing).
- The tonsillectomy guideline position.

- Why steroids before a lymph node biopsy can mislead.

5.2 Common pitfalls

COMMON PITFALLS

- Giving dexamethasone to a child with a suspected lymphoma before the biopsy.
- Forgetting glucose checks in diabetics.

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